실측자료 반영본: 제출 전 확인 사항

반영한 자료. 2026년 10월 1일 실험의 handwritten working log와 IR 출력물, Bruker NMR 원자료를 반영했습니다. 본문 결과 표, 초록, 고찰, 결론을 갱신하고 스펙트럼과 원기록을 부록에 넣었습니다.

질량과 계산 결과

- 4-Bromobenzoic acid 0.3502 g, phenylboronic acid 0.2491 g.
- 빈 봉투 0.8340 g, 봉투+생성물 0.9502 g: 생성물 0.1162 g.
- 이론 생성물 0.3453 g, 분리수율 33.6% (반올림 전 33.6497%).
- 매뉴얼의 Pd 용액 0.25 mM, 1.00 mL와 반응 시간 30 min을 기준으로 하면 Pd loading 0.01435 mol%, TON 2.34×10^{3}, 평균 TOF 78.2 min^{-1}입니다. 실제 촉매 투입량 및 시간의 별도 기입이 없어 이 값들은 **매뉴얼 조건을 전제로 한 계산값**으로 표시했습니다.

분석 결과

- NMR 조건은 **500.13 MHz, $\mathbf{CDCl_3}$**입니다. 잔류 $CHCl_3$ 신호를 7.26 ppm으로 맞추어 전체 축을 -0.023795 ppm 이동했습니다. 원자료는 그대로 보존했습니다.
- 방향족 적분비는 a를 2.00으로 정규화했을 때 2.00:2.04:2.07:2.05:1.00입니다. 주성분의 구조 배정을 뒷받침하지만, 소량의 추가 신호가 있어 정량 순도를 주장하지 않았습니다.
- IR의 넓은 O--H 흡수와 1674 cm^{-1}의 C=O 흡수를 반영했습니다. COOH proton은 제공된 NMR에서 뚜렷하게 분리되지 않습니다.

제출 전에 확인할 사항

1. 질량 중 덧쓴 부분이 있어 위 전사값을 원기록과 대조하십시오. 이 값이 달라지면 수율도 바뀝니다.
2. 기록의 ``0--1?"은 pH paper의 대략적인 판독으로만 반영했습니다. Ethanol을 추가했다는 기록은 있으나 총량과 세척량은 없습니다. 기재되지 않은 색, 결정 형태 및 기기 측정 방식을 만들지 않았습니다.
3. 실험 절차 중 매뉴얼에서 가져온 부피, 시간 및 건조 조건이 실제 수행과 같은지 확인하십시오. 촉매 성능값도 이 확인에 따릅니다.
4. 첨부된 문서는 **working log**이며, 하단의 공식 서명란은 비어 있습니다. 매뉴얼 p. 7의 **TA 서명 official datasheet**는 별도로 확인하여 첨부해야 합니다.

파일 사용. 전체 ZIP에는 원자료, 분석 코드, 그림, 폰트와 보조 LaTeX 패키지가 들어 있습니다. XeLaTeX로 두 번 컴파일합니다. `\draftfalse`로 바꾸면 이 안내 페이지만 숨겨집니다. 공식 datasheet를 `assets/datasheet.pdf`로 넣으면 부록에 추가됩니다.

Aqueous Suzuki–Miyaura Cross-Coupling: Synthesis of 4-Phenylbenzoic Acid

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CM3191 Chemical Synthesis Experiments, Organic Experiment 1

Thursday laboratory, fume hood B7 • Date of experiment: 1 October 2026

Instructor: Dr Hoang Truong Giang

Abstract. The aqueous Suzuki–Miyaura coupling of 4-bromobenzoic acid with phenylboronic acid afforded 0.1162 g of isolated material (33.6% yield) after acidification and recrystallisation from aqueous hydrochloric acid and ethanol. The principal ^{1}H NMR signals (500 MHz, $CDCl_3$) and IR absorptions support assignment of the major component as 4-phenylbenzoic acid. Minor additional NMR signals indicate that analytical purity cannot be assumed. Additional ethanol use during recrystallisation was recorded and may have contributed to product loss. Using the manual's nominal Pd charge and 30 min reaction time gives a loading of 0.0144 mol% Pd, an isolated-mass-based TON of 2.34×10^3 and an average TOF of 78.2 min^{-1}.

1. Introduction

The Suzuki–Miyaura reaction forms a carbon–carbon bond between an organoboron reagent and an organic halide or triflate.[1,2] Boronic acids are useful coupling partners because many are readily handled and compatible with a range of functional groups, although their stability depends on their structure and the reaction conditions.[3] This experiment couples 4-bromobenzoic acid with phenylboronic acid in water, then isolates the biaryl carboxylic acid by acidification and recrystallisation.[6,8]

The conventional mechanism comprises oxidative addition of the aryl bromide to Pd(0), transmetalation of the phenyl group from boron to palladium, and reductive elimination to release the biaryl product and regenerate Pd(0).[2] The catalyst solution is prepared from a Pd(II) precursor. Its entry into this conventional cycle requires activation to a Pd(0) species, but the activation pathway and the identity of the active species are not established by this experiment.

Base affects both substrate solubility and transmetalation. It can produce a trihydroxyboronate from the boronic acid, or promote replacement of a Pd-bound halide by hydroxide, allowing an arylpalladium hydroxide to react with neutral boronic acid. Carrow and Hartwig supported the latter, oxo-palladium pathway for the complexes and weak-base aqueous solvent mixtures they studied.[4] Figure 1 uses that pathway as a simplified illustration; it does not establish the dominant pathway in the present pyrimidine–Pd system.

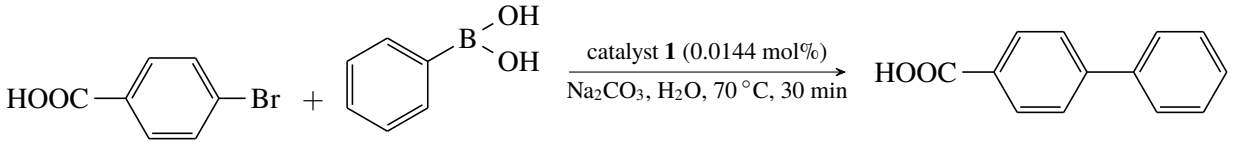


Figure 1: Target transformation (top) and an illustrative Pd(0)/Pd(II) cycle via the oxo-palladium pathway (bottom). The Pd loading uses the nominal catalyst charge. The neutral acid is shown as the isolated product; carboxylate predominates under the basic reaction conditions. Ar = 4-carboxylatophenyl; catalyst **1** = $Pd(OAc)_2$/2-amino-4,6-dihydroxypyrimidine (1:2) in aqueous NaOH. The lower scheme omits precursor activation, ligand exchange and inorganic by-products.

Water can replace an organic reaction solvent when the substrates and catalyst are sufficiently soluble.[5] In this system, the carboxylic acid group provides an ionisable substituent, while the boronic acid participates

in a pH-dependent boronic acid/boronate equilibrium. A phosphine-free, water-soluble catalyst is prepared from $Pd(OAc)_2$ and 2-amino-4,6-dihydroxypyrimidine.[7] The published procedure operates in air and avoids liquid–liquid extraction, although ethanol is still required for purification.[6]

The aims were to synthesise and isolate 4-phenylbenzoic acid, assess its identity and organic impurity profile by IR and ^{1}H NMR spectroscopy, and calculate the isolated yield, TON and average TOF from the measured product mass and total Pd charge.

2. Experimental

2.1 General and record basis

4-Bromobenzoic acid, phenylboronic acid, aqueous Na_2CO_3 (0.5 M), hydrochloric acid (1 M) and ethanol were used as supplied. The manual specifies a catalyst stock containing 0.25 mM Pd, prepared by heating $Pd(OAc)_2$ (5.6 mg, 0.025 mmol), 2-amino-4,6-dihydroxypyrimidine (6.3 mg, 0.050 mmol) and NaOH (4.0 mg, 0.10 mmol) in water (10 mL) at 60 °C for 20 min, then diluting to 100 mL.[8]

The substrate and bag masses below were transcribed from the supplied working log. Unless identified as a recorded value, solution volumes, temperatures and durations reproduce the manual procedure; the actual-value and timing fields on the log were not completed. The noted deviation during recrystallisation is reported explicitly. The quantitative calculations use the measured masses and, where required, the nominal catalyst charge and reaction time.

The IR printout was generated with PerkinElmer Spectrum IR software, version 10.6.1; the instrument model and sampling mode were not recorded. The Bruker ^{1}H NMR dataset specifies $CDCl_3$, 500.13 MHz, eight scans, the `zg30` pulse programme and a 1.0 s relaxation delay. The supplied processed spectrum was retained, with its chemical-shift axis shifted by -0.023795 ppm to place residual $CHCl_3$ at δ 7.26.[9] Aromatic integrals were obtained by numerical integration of this spectrum and normalised to signal a = 2.00. This routine acquisition was not validated for quantitative purity analysis.

2.2 Synthesis and purification

4-Bromobenzoic acid (0.3502 g, 1.742 mmol, 1.00 equiv) and phenylboronic acid (0.2491 g, 2.043 mmol, 1.173 equiv) were charged to a stirred reaction vial. The manual procedure calls for aqueous Na_2CO_3 (0.5 M, 10 mL, 5.0 mmol), stirring until dissolution, addition of catalyst solution (0.25 mM, 1.0 mL, 0.250 μmol Pd), and heating at 70 °C for 30 min. The log records no visible Pd black after this heating stage.

The prescribed workup is to cool the mixture to room temperature, transfer it to a 250 mL flask, and add 1 M HCl (20 mL) dropwise with stirring in an ice bath. The handwritten pH-paper estimate was approximately 0–1, with a question mark indicating uncertainty. The protocol then specifies water (20 mL), a further 10 min in the ice bath, and collection of the solid by suction filtration. Gas evolution and precipitate appearance were not entered as actual observations in the log.

For recrystallisation, the manual specifies 1 M HCl (3 mL) and ethanol (10 mL), with the minimum additional hot ethanol needed for dissolution. The log notes longer boiling and unintended additional ethanol use; neither the total ethanol volume nor the wash volume was recorded. The specified cooling sequence is room temperature for 10–15 min, then an ice bath for 15 min, followed by suction filtration, a cold-ethanol wash and 15 min drying under an IR lamp. The recovered solid gave a bag-plus-product mass of 0.9502 g; subtraction of the empty-bag mass (0.8340 g) gave 0.1162 g (0.5862 mmol, 33.6%). Constant mass and the dried solid's colour and form were not documented. The NMR sample was measured in $CDCl_3$; its exact sample mass and solvent volume were not recorded.

Characterisation. IR $\tilde{\nu}$/cm^{-1}: ca. 3300–2500 (br, acid O–H), 1674, 1607, 1487, 1449, 1316, 1299, 1286, 861, 744, 695. ^{1}H NMR (500 MHz, $CDCl_3$) δ: 8.19 (d, $J = 8.3$ Hz, 2H, a), 7.71 (d, $J = 8.3$ Hz, 2H, b), 7.66–7.64 (m, 2H, c), 7.50–7.47 (m, 2H, d), 7.43–7.40 (m, 1H, e). Proton counts describe the structural assignments; the measured relative integrals are given in Table 4. A distinct COOH resonance was not resolved.

3. Results

Table 1: Reagent quantities and their evidential basis. Equivalents are relative to 4-bromobenzoic acid.

Reagent	Amount	n/mmol	Equiv.	Basis
4-Bromobenzoic acid	0.3502 g	1.742	1.000	Recorded
Phenylboronic acid	0.2491 g	2.043	1.173	Recorded
Na_2CO_3 (0.5 M)	10 mL	5.0	2.87	Protocol
Pd solution (0.25 mM)	1.0 mL	0.000250	0.000144	Protocol

Table 2: Mass balance, isolated yield and nominal Pd-based performance.

Quantity	Value
Empty plastic bag	0.8340 g
Bag and isolated product	0.9502 g
Isolated product mass (difference)	0.1162 g
Theoretical product mass (4-bromobenzoic acid limiting)	0.3453 g
Isolated product amount	0.5862 mmol
Isolated yield	33.6%
Nominal total Pd charge	0.000250 mmol
Nominal Pd loading	0.01435 mol%
Isolated-mass-based TON	2.34×10^3
Average TOF over the nominal 30 min reaction	78.2 min^{-1}

For the 1:1 coupling, 4-bromobenzoic acid is limiting: $0.3502/201.02 = 1.742 \times 10^{-3}$ mol, compared with $0.2491/121.93 = 2.043 \times 10^{-3}$ mol of phenylboronic acid. Thus,

$$
\begin{aligned}
m_{\text{iso}} &= 0.9502 - 0.8340 = 0.1162 \text{ g},\\
m_{\text{theor}} &= (0.3502/201.02) \times 198.22 = 0.345322 \text{ g},\\
\text{yield} &= 100 \times 0.1162/0.345322 = 33.6\%,\\
\text{TON} &= \frac{0.1162/198.22}{(0.25 \times 10^{-3})(1.00 \times 10^{-3})} = 2345,\\
\overline{\text{TOF}} &= 2345/30 = 78.2 \text{ min}^{-1}.
\end{aligned}
$$

These are gravimetric estimates assuming the recovered mass is dry product. The TON and average TOF additionally assume the protocol catalyst charge and reaction duration; they are not independent measurements of catalyst activity.

Table 3: Actual handwritten observations, distinguished from the printed expected-observation prompts.

Stage	Recorded observation
After heating	No visible Pd black solids.
Acidification	pH-paper estimate approximately 0–1; recorded with a question mark.
Recrystallisation	Longer boiling and unintended additional ethanol use; amount not recorded.
Other stages	No actual entry for initial dissolution, gas evolution, crystal appearance or dried product colour.

Table 4: Measured ^{1}H NMR data in $CDCl_3$. Letters refer to the structure in Section 4.3.

H	δ/ppm	Mult., J/Hz	Exp. H	Rel. int.	Assignment
a	8.19	d, 8.3	2	2.00	*ortho* to CO_2H
b	7.71	d, 8.3	2	2.04	acid-bearing ring, *ortho* to Ph
c	7.66–7.64	m	2	2.07	terminal phenyl, *ortho*
d	7.50–7.47	m	2	2.05	terminal phenyl, *meta*
e	7.43–7.40	m	1	1.00	terminal phenyl, *para*
COOH	Not resolved	—	1	—	exchangeable acid proton

Integration limits (high to low ppm): a 8.250–8.140; b 7.760–7.675; c 7.675–7.600; d 7.540–7.455; e 7.455–7.350. Relative areas use a = 2.00; they were not forced to sum to 9. Multiplets c–e are not assigned artificial first-order coupling constants.

Table 5: Principal observed IR absorptions and assignments. Positions are rounded from the supplied printout.

Observed / cm^{-1}	Assignment[11]
ca. 3300–2500 (broad)	Hydrogen-bonded carboxylic acid O–H stretch
1674 (strong)	Conjugated carboxylic acid C=O stretch
1607, 1487, 1449	Aromatic ring stretching bands
1316, 1299, 1286	Carboxylic acid C–O stretching region, with coupled modes
861	Consistent with a *para*-disubstituted aromatic ring
744, 695	Consistent with a monosubstituted phenyl ring

4. Discussion

4.1 Why the coupling works in water

The carboxylic acid group makes the aryl bromide compatible with the aqueous medium through formation of its carboxylate salt. Phenylboronic acid also participates in the equilibrium

$$\mathrm{PhB(OH)_2 + OH^- \rightleftharpoons [PhB(OH)_3]^-}.$$

These acid–base equilibria favour dissolution under the basic conditions, while the water-soluble Pd–pyrimidine catalyst provides access to a catalytically competent palladium species.[7] Dissolving the substrates before catalyst addition reduces complications from undissolved starting material.

The pH of the carbonate stock solution cannot be assigned directly to the reaction mixture. Addition of the carboxylic acid and equilibration of the boronic acid consume basic equivalents and change the carbonate/bicarbonate distribution. Consequently, exact carboxylate or boronate fractions are not calculated from the nominal 0.5 M carbonate concentration. Base also facilitates transmetalation and neutralises acidic products. A formal net equation, using one choice of acid–base species, is

$$\mathrm{Br{-}C_6H_4{-}CO_2^- + PhB(OH)_2 + OH^- \longrightarrow Ph{-}C_6H_4{-}CO_2^- + B(OH)_3 + Br^-}.$$

The equation expresses the coupling stoichiometry; it is not a complete description of the solution speciation.

This procedure benefits from an ionisable substrate and a water-compatible catalyst, so its performance cannot be assumed for an insoluble, non-ionisable aryl halide. The 1.173 equivalents of phenylboronic acid provide an excess coupling partner. Protodeboronation or homocoupling could consume some of that excess, but neither process is established without supporting analytical evidence.[3]

4.2 Acid–base control of isolation and purification

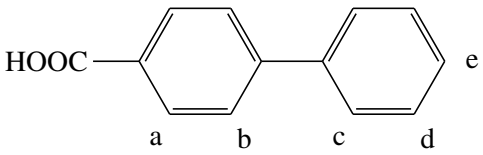
Acidification protonates the product carboxylate and neutralises carbonate and bicarbonate, releasing carbon dioxide:

$$CO_3^{2-} + 2\,H^+ \longrightarrow CO_2 + H_2O, \qquad ArCO_2^- + H^+ \longrightarrow ArCO_2H.$$

The intended HCl charge is in excess of the basic equivalents introduced, and pH paper provides an experimental check that the final mixture is acidic. The recorded pH-paper estimate of approximately 0–1 supports acidification, but its question mark and the limited resolution of pH paper preclude an exact pH assignment. Protonation favours the less soluble neutral acid, while cooling promotes precipitation and limits the temperature rise during acid addition. The aqueous phase retains soluble inorganic salts and boron-containing material. Some neutral product can nevertheless remain dissolved, so protonation does not imply quantitative recovery.

Recrystallisation uses the higher solubility of the product in hot aqueous ethanol than in the cold mixture. The added HCl maintains acidic conditions. Using the minimum volume of hot ethanol limits the amount of product retained in the cold mother liquor; slow, undisturbed cooling allows crystal growth before the final ice-bath cooling. Cold ethanol removes adhering mother liquor during washing but can also dissolve product. Unreacted aryl acid or organic by-products may be retained with the crystals, so purification cannot be inferred from their appearance alone. The recorded additional ethanol use is therefore relevant to product recovery, although the absence of a measured volume prevents a quantitative estimate of the associated loss.

4.3 Identity and purity of the product



The para-disubstituted ring gives two apparent doublets at δ 8.19 (a) and 7.71 (b), each with a resolved splitting of approximately 8.3 Hz. The terminal phenyl ring accounts for the multiplets at δ 7.66–7.64 (c), 7.50–7.47 (d) and 7.43–7.40 (e). Their relative integrals, with a = 2.00, are 2.00:2.04:2.07:2.05:1.00, close to the structural ratio 2:2:2:2:1. The assignments agree with reported $CDCl_3$ data for 4-phenylbenzoic acid: δ 8.19 (d, $J = 8.2$ Hz, 2H), 7.71 (d, $J = 8.2$ Hz, 2H), 7.64 (d, $J = 7.8$ Hz, 2H) and 7.55–7.38 (m, 3H).[10] The current spectrum resolves the last three protons into two groups; c–e are conservatively reported as multiplets.

This assignment rests on the complete pattern, including the downfield 2H doublet and the relative proton counts. A monosubstituted phenyl pattern alone would not establish coupling, because phenylboronic acid and biphenyl also contain such a ring. A distinct COOH signal was not resolved in the supplied spectrum, including an expanded view of δ 9–14. Exchange can weaken or broaden this resonance, so its absence does not contradict the acid assignment and it was not used as an integration reference.

Small non-product signals are visible. A singlet at δ 2.175 is consistent with acetone, and a weak quartet near 3.73 with a triplet near 1.25 is consistent with ethanol.[9] These assignments remain tentative because a chemical shift is not unique identification. Weak additional signals near 4.41 and 1.42, and low-intensity aromatic features, were not conclusively assigned. The source of these traces cannot be separated between the isolated material and sample preparation from one spectrum. The principal aromatic pattern supports

4-phenylbenzoic acid as the major component, but no numerical purity is claimed. The 1.0 s relaxation delay was not validated against proton relaxation times, and this spectrum does not assay inorganic salts or residual palladium. Nor does the purified sample establish reaction conversion before workup.

The broad IR absorption over approximately 3300–2500 cm^{-1} and the strong band at 1674 cm^{-1} support a hydrogen-bonded carboxylic acid with a conjugated carbonyl group. These features alone cannot distinguish product from 4-bromobenzoic acid. Bands at 861, 744 and 695 cm^{-1} provide supporting evidence for the para-disubstituted and monosubstituted aromatic rings. Together, the IR and complete NMR pattern support the proposed product structure, while the minor NMR signals qualify the purity assessment.

4.4 Yield

The isolated yield was 33.6%, corresponding to 0.1162 g recovered against a theoretical 0.3453 g. This is an overall isolated yield, combining product formation with losses during transfer, precipitation, filtration and recrystallisation; it is neither reaction conversion nor a separately measured recrystallisation recovery.

The working log records longer boiling and unintended additional ethanol use. A larger solvent volume can leave more product dissolved after cooling, so recrystallisation loss is a plausible contributor to the low isolated yield. Its extent cannot be calculated because the total ethanol volume, crude product mass and mother-liquor composition were not measured. Transfer and filtration losses, or incomplete reaction, may also contribute. No visible Pd black was reported, but this observation does not establish that all Pd remained molecularly dispersed or catalytically active. It therefore does not exclude catalyst deactivation or prove complete conversion.

Conversely, residual solvent or other impurities can inflate the weighed mass. The trace NMR signals warrant this qualification, although they do not determine the bulk impurity mass fraction. The protocol's 15 min IR-lamp drying period is not evidence of constant mass. The most useful procedural improvements would be to record the total recrystallisation and wash volumes, add hot ethanol only as needed for dissolution, minimise transfer losses, and verify constant mass where the course procedure permits.

4.5 Catalyst performance and green chemistry

Using the nominal 0.250 μmol Pd charge gives 0.01435 mol% relative to the measured aryl bromide amount. The isolated mass gives a TON of 2.34×10^3 and an average TOF of 78.2 min^{-1} over the nominal 30 min reaction (Table 2). These estimates require confirmation that the protocol catalyst volume and reaction duration were used, because the corresponding actual-value fields are blank in the log. They are normalised to all Pd added, not to a measured active-site concentration.[14] Isolation losses can reduce the calculated TON relative to product formation, whereas contamination can raise a mass-derived value. A single isolated-product measurement cannot determine an initial TOF, catalyst activation time or the time at which product formation stopped.

Water is the reaction solvent, the reported procedure can be performed in air, and product isolation avoids liquid–liquid extraction.[6] These features reduce organic solvent demand during reaction and workup. Purification still consumes ethanol, however, and heating, cooling, excess base and acid, and palladium-containing waste remain part of the process. Low Pd loading does not establish catalyst recovery or reuse.

Atom economy concerns the balanced stoichiometric reaction rather than the experimental yield.[12] Dividing the product molar mass by the sum of the two coupling-partner molar masses gives $198.22/(201.02 + 121.93) = 61.4\%$, but this restricted calculation omits the stoichiometric base and is not a complete atom-economy assessment. A process-level mass metric must also account for actual reagent and solvent use, including purification.[13,15] An E-factor would require the actual amounts consumed and an explicit statement of whether water is included. The aqueous filtrate and ethanol mother liquor are the main liquid waste streams; their disposal follows the laboratory's designated waste procedures.

5. Conclusion

The aqueous Suzuki–Miyaura coupling afforded 0.1162 g of isolated material in 33.6% yield. Its principal aromatic NMR pattern and IR bands support assignment of the major component as 4-phenylbenzoic acid, while minor additional NMR signals prevent a claim of analytical purity. Recorded additional ethanol use may have lowered recovery during recrystallisation, but the available data do not distinguish workup loss from incomplete reaction. On the nominal catalyst charge and 30 min reaction time, the isolated-mass-based TON and average TOF are 2.34×10^3 and 78.2 min^{-1}, respectively. The experiment demonstrates coupling in water through substrate ionisation and a compatible catalyst, with product recovery and purification solvent use remaining important to the overall outcome.

References

[1] Miyaura, N.; Yamada, K.; Suzuki, A. A New Stereospecific Cross-Coupling by the Palladium-Catalyzed Reaction of 1-Alkenylboranes with 1-Alkenyl or 1-Alkynyl Halides. *Tetrahedron Lett.* **1979**, *20*, 3437–3440.

[2] Miyaura, N.; Suzuki, A. Palladium-Catalyzed Cross-Coupling Reactions of Organoboron Compounds. *Chem. Rev.* **1995**, *95*, 2457–2483.

[3] Lennox, A. J. J.; Lloyd-Jones, G. C. Selection of Boron Reagents for Suzuki–Miyaura Coupling. *Chem. Soc. Rev.* **2014**, *43*, 412–443.

[4] Carrow, B. P.; Hartwig, J. F. Distinguishing Between Pathways for Transmetalation in Suzuki–Miyaura Reactions. *J. Am. Chem. Soc.* **2011**, *133*, 2116–2119.

[5] Anastas, P. T.; Warner, J. C. *Green Chemistry: Theory and Practice*; Oxford University Press: Oxford, 1998.

[6] Hamilton, A. E.; Buxton, A. M.; Peeples, C. J.; Chalker, J. M. An Operationally Simple Aqueous Suzuki–Miyaura Cross-Coupling Reaction for an Undergraduate Organic Chemistry Laboratory. *J. Chem. Educ.* **2013**, *90*, 1509–1513.

[7] Chalker, J. M.; Wood, C. S. C.; Davis, B. G. A Convenient Catalyst for Aqueous and Protein Suzuki–Miyaura Cross-Coupling. *J. Am. Chem. Soc.* **2009**, *131*, 16346–16347.

[8] Hoang, T. G. *CM3191 Chemical Synthesis Experiments: Organic Lab Manual*; National University of Singapore, AY2026/27; Experiment 1, pp 2–7.

[9] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[10] Peramo, A.; Abdellah, I.; Pecnard, S.; Mougin, J.; Martini, C.; Couvreur, P.; Huc, V.; Desmaële, D. A Self-Assembling NHC-Pd-Loaded Calixarene as a Potent Catalyst for the Suzuki–Miyaura Cross-Coupling Reaction in Water. *Molecules* **2020**, *25*, 1459; characterisation of compound 3e. `https://doi.org/10.3390/molecules25061459`.

[11] Pavia, D. L.; Lampman, G. M.; Kriz, G. S.; Vyvyan, J. R. *Introduction to Spectroscopy*, 5th ed.; Cengage Learning: Stamford, CT, 2015.

[12] Trost, B. M. The Atom Economy—A Search for Synthetic Efficiency. *Science* **1991**, *254*, 1471–1477.

[13] Sheldon, R. A. The E Factor: Fifteen Years On. *Green Chem.* **2007**, *9*, 1273–1283.

[14] Bligaard, T.; Bullock, R. M.; Campbell, C. T.; Chen, J. G.; Gates, B. C.; Gorte, R. J.; Jones, C. W.; Jones, W. D.; Kitchin, J. R.; Scott, S. L. Toward Benchmarking in Catalysis Science: Best Practices, Challenges, and Opportunities. *ACS Catal.* **2016**, *6*, 2590–2602. `https://doi.org/10.1021/acscatal.6b00183`.

[15] Constable, D. J. C.; Curzons, A. D.; Cunningham, V. L. Metrics to 'Green' Chemistry—Which Are the Best? *Green Chem.* **2002**, *4*, 521–527. `https://doi.org/10.1039/B206169B`.

A. NMR spectra and processing record

The figures below were generated from the supplied Bruker processed real spectrum (experiment 1, `pdata/1/1r`). The original FID, parameters, peak list and processed data are preserved in the accompanying source archive. No additional phasing, baseline correction, smoothing or removal of impurity peaks was applied. Only a uniform chemical-shift correction and intensity scaling were used for display. The original solvent peak at 7.283795 ppm was set to 7.260000 ppm. The 65 536-point processed spectrum has a digital spacing of 0.153 Hz; the reported doublet couplings were calculated from the supplied peak-list separations using 500.13 MHz.

Integration used the windows stated below Table 4, with a normalised to 2.00. The integral sum is 9.16 rather than a forced 9.00. Shifting all integration windows by ± 0.005 ppm changed any normalised group area by less than 0.01, but this numerical stability does not validate quantitative purity or correct for relaxation effects. The accompanying `analysis/analyse_nmr.py` reproduces the figures and calculations.

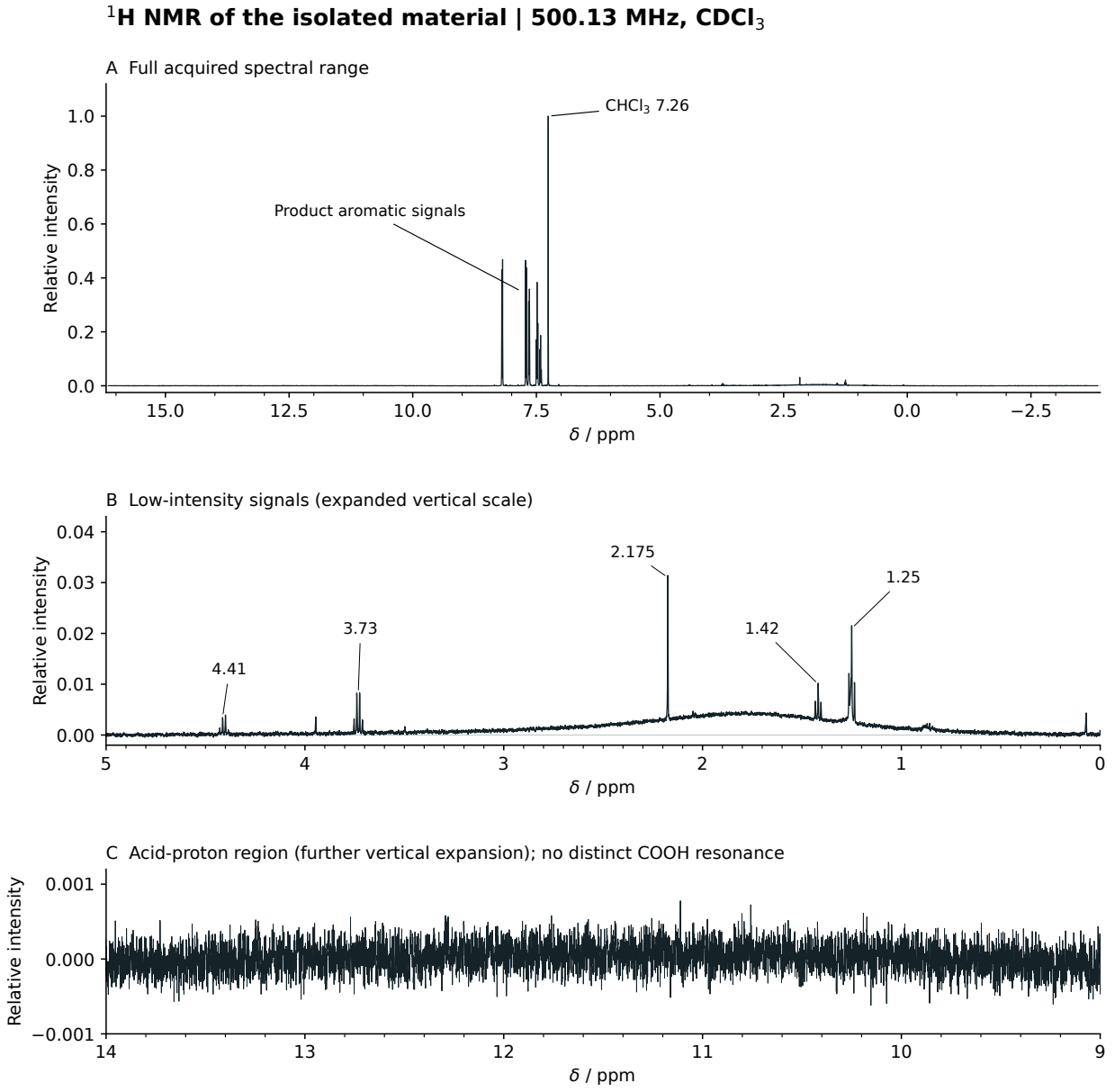


Figure 2: Measured ^{1}H NMR spectrum with expanded views of low-intensity signals and the acid-proton region. Vertical scales differ between panels. Small extra peaks are retained.

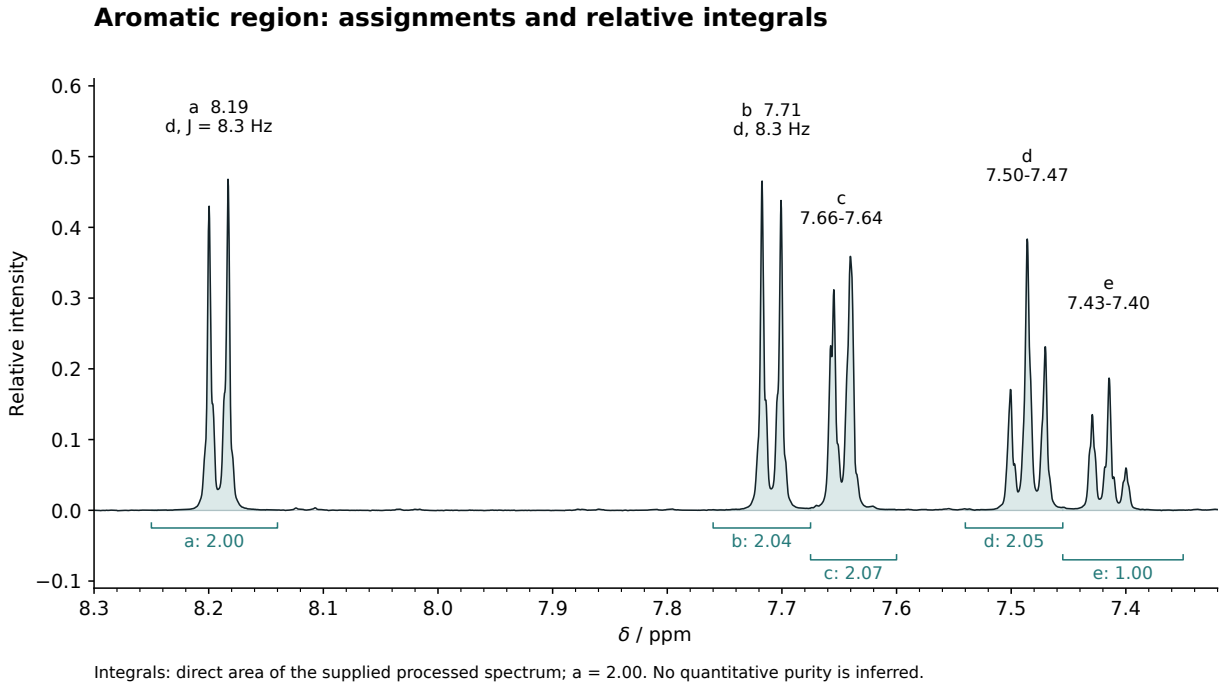


Figure 3: Aromatic-region expansion with assignments a–e, integration windows and measured relative areas. The $CHCl_3$ reference at 7.26 ppm lies outside this enlargement.

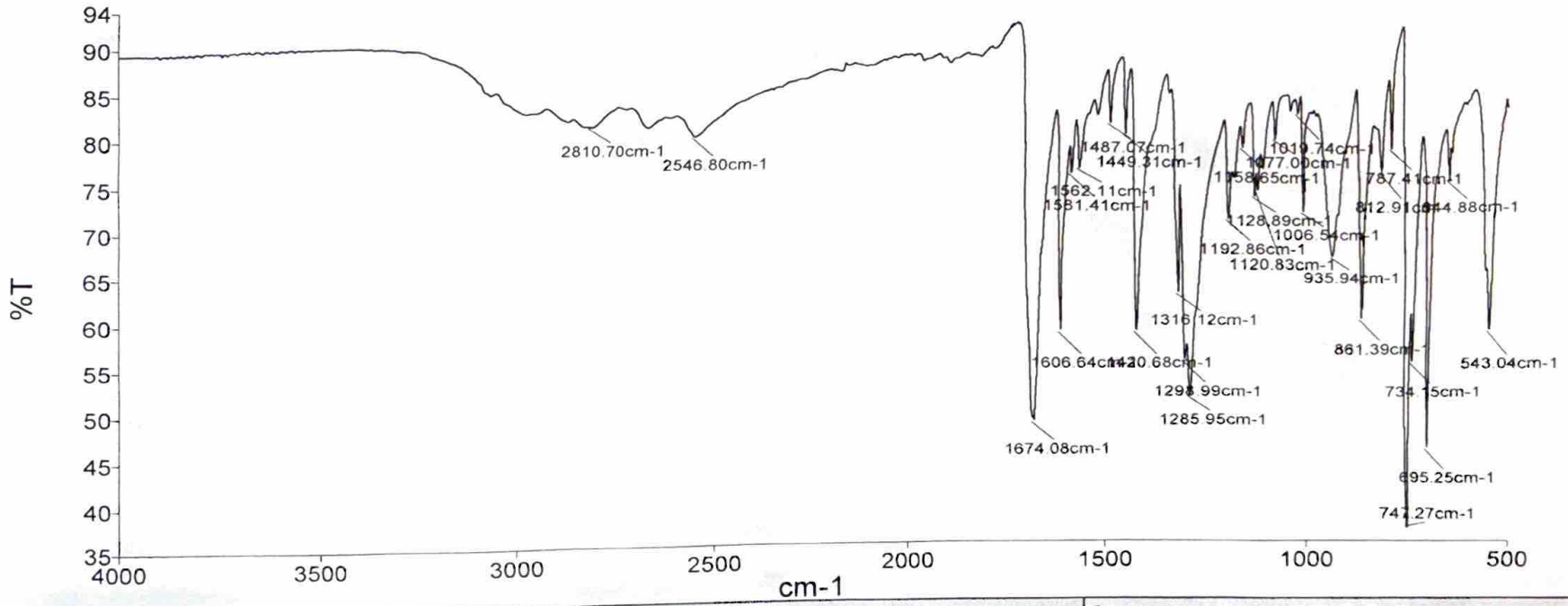
B. Original experimental records

The following pages reproduce the supplied IR printout and annotated working log. The IR page has been rotated for readability; its spectral content has not been altered. The working log includes handwritten masses and observations, but its formal student/GTA signature fields are blank and its footer distinguishes it from the official datasheet. The TA-signed official datasheet required by the manual must therefore be supplied separately.[8]

PerkinElmer Spectrum IR Version 10.6.1
Thursday, 1 October, 2026 2:35 PM

Analyst Administrator

Date Thursday, 1 October, 2026 2:35 PM



Sample Name	Description	Quality Checks
Suzuki_KimDongHwan	Sample 980 By Administrator Date Thursday, October 01 2026	The Quality Checks do not report any warnings for the sample.

DATASHEET — WORKING LOG

LHN

Suzuki–Miyaura Cross-Coupling in Water

Name: KIM DONGHWAN Fumehood: B7 Date: 1 Oct 2026 Photos taken: ☐

1. Masses

4-Bromobenzoic acid (g)	0.3502	Phenylboronic acid (g)	0.2491
n(ArBr) = mass/201.02 (mmol)		Equiv. $PhB(OH)_2$	
Empty plastic bag (g)	0.8340	Bag + product (g)	0.9502
Product (g)		Theoretical (g) = $0.9861\ m_{ArBr}$	
		Percentage yield (%)	

2. Time log

Pd added (clock)	Out of block	HCl addition done	Into IR lamp

3. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Solids + Na_2CO_3, stirred	all reactants dissolve [M]; clear, colourless [E]		
Pd catalyst solution	pale yellow [M, App. 1]		
After 30 min at 70 °C	*record colour and any solid or grey tint (Pd black)* [E]	no Pd black solids	
HCl addition	gas forms [M] (CO_2 [C]); solid precipitates [E]		
pH paper after HCl	acidic [M]; calc. pH ≈ 0.7 [C]	0 ~ 1 ?	
Hot ethanol, solution 1	all material dissolves with ~~minimal ethanol~~ [M]	boiled longer ~> eventually put more EtOH	
Cooling to RT, then ice	crystals form (recrystallisation) [M]	& reason of small yield?	
Dried product	white solid [E]		

Sources: [M] stated in the lab manual; [C] calculated from manual quantities; [E] general expectation, not stated in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see; a mismatch is a result worth discussing.

4. Ethanol used in recrystallization & catalyst metrics

Hot EtOH added from sol. 2 (mL, est.)		n(product) (mmol)	
TON = $n_{product}/(2.5 \times 10^{-4}$ mmol)		TOF = TON/30 min (1/min)	
IR file name		NMR tube no. / solvent	

5. Other observations

To be completed at the end of the session		
Student signature	GTA name	GTA signature

This is a working log. The **official** datasheet (lab manual p. 7) must still be printed, signed by the GTA and submitted with the report.